AMPLENESS OF AN INVERTIBLE SHEAF ON A ONE-DIMENSIONAL PROPER REDUCED SCHEME IS CHECKED BY POSITIVITY ON EACH CURVE COMPONENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The author's parenthetical 'details omitted' concerns only the elementary verification that the coproduct morphism is finite and surjective; the rest of the argument is written out.

Allowed standard prerequisites (editorial).

- Ampleness descends along finite surjective morphisms. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-surjective-finite-morphism-ample (tag 0B5V). Justifies replacing X by the coproduct of its components when checking ampleness.

Source reference. Original reference 0B5V: sources/stacks-chow/tags.txt, source lines 14048–14048. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B5X: sources/stacks-chow/tags.txt, source lines 14050–14050. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a proper scheme of dimension ≤ 1 over k. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Let $C_i \subset X$, $i = 1, \ldots, t$ be the irreducible components of dimension 1. The following are equivalent:*

(1) *$\mathcal{L}$ is ample, and*
(2) *$\deg(\mathcal{L}|_{C_i}) > 0$ for $i = 1, \ldots, t$.*

Proof. Let $x_1, \ldots, x_r \in X$ be the isolated closed points. Think of $x_i = \mathrm{Spec}(\kappa(x_i))$ as a scheme. Consider the morphism of schemes

$$f : C_1 \amalg \ldots \amalg C_t \amalg x_1 \amalg \ldots \amalg x_r \longrightarrow X$$

This is a finite surjective morphism of schemes proper over k (details omitted). Thus $\mathcal{L}$ is ample if and only if $f^*\mathcal{L}$ is ample (Cohomology of Schemes, Lemma 0B5V). Thus we conclude by Lemma 0B5X. □

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